MATH 243: Discussion Worksheet 9 **Answer Key**

Topics: 14.8 Lagrange Multipliers; 15.1 Double Integrals over Rectangles
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1. Use the method of Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint.

   (a) $f(x, y) = 9xe^y$, subject to $x^2 + y^2 = 2$.

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x, y) = 2,$$

where $g(x, y) = x^2 + y^2$. From $\nabla f = \lambda \nabla g$, we get

$$\begin{aligned} f_x &= \lambda g_x \qquad \Rightarrow \qquad 9e^y = 2\lambda x, \\ f_y &= \lambda g_y \qquad \Rightarrow \qquad 9xe^y = 2\lambda y. \end{aligned}$$

So the system is

$$9e^y = 2\lambda x, \qquad 9xe^y = 2\lambda y, \qquad x^2 + y^2 = 2.$$

Multiply the first equation by x to get

$$9xe^y = 2\lambda x^2.$$

Comparing with the second equation, we obtain

$$2\lambda x^2 = 2\lambda y.$$

Since $9e^y = 2\lambda x$ and $e^y \neq 0$, we have $\lambda \neq 0$. Hence

$$x^2 = y.$$

Substituting into the constraint,

$$x^2 + (x^2)^2 = 2 \quad \Rightarrow \quad x^4 + x^2 - 2 = 0 \quad \Rightarrow \quad (x^2 + 2)(x^2 - 1) = 0.$$

Since $x^2 + 2 > 0$ for every real x, we must have $x^2 - 1 = 0$. Thus $x = \pm 1$, and since $y = x^2$, we have $y = 1$.
So the candidate points are $(1, 1)$ and $(-1, 1)$. Evaluating:

$$f(1, 1) = 9e, \qquad f(-1, 1) = -9e.$$

Therefore, the maximum value is $9e$ at $(1, 1)$, and the minimum value is $-9e$ at $(-1, 1)$.

   (b) $f(x, y) = 81x^2 + y^2$, subject to $4x^2 + y^2 = 9$.

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x, y) = 9,$$

where $g(x,y) = 4x^2 + y^2$. From $\nabla f = \lambda \nabla g$, we get

$$\begin{aligned} f_x &= \lambda g_x \qquad \Rightarrow \qquad 162x = 8\lambda x, \\ f_y &= \lambda g_y \qquad \Rightarrow \qquad 2y = 2\lambda y. \end{aligned}$$

So the system is

$$162x = 8\lambda x, \qquad 2y = 2\lambda y, \qquad 4x^2 + y^2 = 9.$$

Rearranging and factoring the second equation,

$$2y(1 - \lambda) = 0,$$

so either $y = 0$ or $\lambda = 1$.
If $y = 0$, then the constraint gives

$$4x^2 = 9 \quad \Rightarrow \quad x = \pm\frac{3}{2}.$$

If $\lambda = 1$, then the first equation gives

$$162x = 8x \quad \Rightarrow \quad 154x = 0 \quad \Rightarrow \quad x = 0.$$

Substituting into the constraint,

$$y^2 = 9 \quad \Rightarrow \quad y = \pm 3.$$

Thus the candidate points are

$$\left(\frac{3}{2}, 0\right), \quad \left(-\frac{3}{2}, 0\right), \quad (0, 3), \quad (0, -3).$$

Evaluating:

$$\begin{aligned} f\left(\frac{3}{2}, 0\right) &= \frac{729}{4}, \qquad f\left(-\frac{3}{2}, 0\right) = \frac{729}{4}, \\ f(0, 3) &= 9, \qquad f(0, -3) = 9. \end{aligned}$$

Therefore, the maximum value is $\frac{729}{4}$ at $\left(\pm\frac{3}{2}, 0\right)$, and the minimum value is 9 at $(0, \pm 3)$.

2. A rectangle in $\mathbb{R}^2$ has one vertex at the origin, two sides along the positive coordinate axes, and its opposite vertex in the first quadrant on the curve $x^3 + y^3 = 16$. Use the method of Lagrange multipliers to find the dimensions of the rectangle with the largest area.

Let (x, y) be the vertex of the rectangle on the curve, where $x, y > 0$. The side lengths are x and y. The area of the rectangle is $f(x, y) = xy$.

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x, y) = 16,$$

where $g(x, y) = x^3 + y^3$.

From $\nabla f = \lambda \nabla g$, we get

$$y = 3\lambda x^2, \qquad x = 3\lambda y^2, \qquad x^3 + y^3 = 16.$$

Multiplying the first equation by x and the second by y, we get

$$xy = 3\lambda x^3 = 3\lambda y^3.$$

Since $x, y > 0$, the first equation gives $\lambda = y/(3x^2) > 0$. We can therefore divide by 3λ to obtain $x^3 = y^3$. Thus, $x = y$.

Substituting into the constraint,

$$2x^3 = 16 \quad \Rightarrow \quad x = 2.$$

Then $y = 2$.

For comparison, include the endpoints $(0, \sqrt[3]{16})$ and $(\sqrt[3]{16}, 0)$ of the arc in the closed first quadrant. At either endpoint, the area is 0. The area is continuous on this closed and bounded arc, so it attains a maximum. Since $f(2,2) = 4 > 0$, the maximum occurs at $(2,2)$.

Therefore, the dimensions of the rectangle with the largest area are 2×2, and the maximum area is 4.

3. Find the absolute maximum and absolute minimum values of the function

$$f(x,y) = 2x^2 - y^2 + 6y$$

on the disk $x^2 + y^2 \leq 16$.

Since f is continuous and $D = \{(x,y) : x^2 + y^2 \leq 16\}$ is closed and bounded, the absolute maximum and absolute minimum must occur either at a critical point in the interior of D or on the boundary of D.

We compute

$$f_x(x,y) = 4x, \qquad f_y(x,y) = -2y + 6.$$

Setting these equal to zero gives

$$4x = 0, \qquad -2y + 6 = 0,$$

so the only critical point is $(0,3)$, which lies in the interior of D. At this point,

$$f(0,3) = 2(0)^2 - 3^2 + 6(3) = 9.$$

Next we check the boundary $x^2 + y^2 = 16$.

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x,y) = 16,$$

where $g(x,y) = x^2 + y^2$. From $\nabla f = \lambda \nabla g$, we get

$$\begin{aligned} f_x = \lambda g_x \quad &\Rightarrow \quad 4x = 2\lambda x, \\ f_y = \lambda g_y \quad &\Rightarrow \quad -2y + 6 = 2\lambda y. \end{aligned}$$

So the system is

$$4x = 2\lambda x, \qquad -2y + 6 = 2\lambda y, \qquad x^2 + y^2 = 16.$$

Rearranging and factoring the first equation,

$$2x(2 - \lambda) = 0,$$

so either $x = 0$ or $\lambda = 2$.

If $x = 0$, then the constraint gives

$$y^2 = 16 \quad \Rightarrow \quad y = \pm 4.$$

If $\lambda = 2$, then the second equation gives

$$-2y + 6 = 4y \quad \Rightarrow \quad 6y = 6 \quad \Rightarrow \quad y = 1.$$

Substituting into the constraint,

$$x^2 + 1 = 16 \quad \Rightarrow \quad x^2 = 15 \quad \Rightarrow \quad x = \pm\sqrt{15}.$$

Thus the candidate points on the boundary are

$$(0, 4), \quad (0, -4), \quad (\sqrt{15}, 1), \quad (-\sqrt{15}, 1).$$

Evaluating:

$$f(0, 4) = 8, \qquad f(0, -4) = -40,$$
$$f(\sqrt{15}, 1) = 35, \qquad f(-\sqrt{15}, 1) = 35.$$

Comparing these values with the value $f(0, 3) = 9$ at the interior critical point, we conclude that the absolute maximum value is 35 at $(\pm\sqrt{15}, 1)$, and the absolute minimum value is -40 at $(0, -4)$.

4. Evaluate each double integral.

(a) $\displaystyle\iint_R \left(y + \frac{x}{y^2}\right) dA$, where $R = \{(x, y) \mid 0 \le x \le 2,\ 1 \le y \le 2\}$.

$$\begin{aligned}
\iint_R \left(y + \frac{x}{y^2}\right) dA &= \int_0^2 \int_1^2 \left(y + \frac{x}{y^2}\right) dy\, dx \\
&= \int_0^2 \left[\frac{y^2}{2} - \frac{x}{y}\right]_{y=1}^{y=2} dx \\
&= \int_0^2 \left[\left(2 - \frac{x}{2}\right) - \left(\frac{1}{2} - x\right)\right] dx \\
&= \int_0^2 \left(\frac{3}{2} + \frac{x}{2}\right) dx \\
&= \left[\frac{3x}{2} + \frac{x^2}{4}\right]_0^2 \\
&= 4.
\end{aligned}$$

(b) $\displaystyle\iint_R x\cos(xy)\, dA$, where $R = [0, \pi] \times [0, 1]$.

$$\begin{aligned}\iint_R x\cos(xy)\,dA &= \int_0^{\pi}\int_0^{1} x\cos(xy)\,dy\,dx \\ &= \int_0^{\pi} \left[\sin(xy)\right]_{y=0}^{y=1}\,dx \\ &= \int_0^{\pi} \sin x\,dx \\ &= \left[-\cos x\right]_0^{\pi} \\ &= 2.\end{aligned}$$

(c) $\displaystyle\iint_R \frac{1}{(2x+3y)^2}\,dA$, where R is the rectangle with vertices $(0,1),\ (1,1),\ (1,2),\ (0,2)$.

The vertices give the bounds $0 \le x \le 1$ and $1 \le y \le 2$.

$$\begin{aligned}\iint_R \frac{1}{(2x+3y)^2}\,dA &= \int_1^{2}\int_0^{1} (2x+3y)^{-2}\,dx\,dy \\ &= \int_1^{2} \left[-\frac{1}{2(2x+3y)}\right]_{x=0}^{x=1}\,dy \\ &= -\frac{1}{2}\int_1^{2} \left(\frac{1}{2+3y} - \frac{1}{3y}\right)\,dy \\ &= -\frac{1}{6}\left[\ln(2+3y) - \ln y\right]_1^2 \\ &= -\frac{1}{6}\big(\ln 8 - \ln 2 - \ln 5 + \ln 1\big) \\ &= -\frac{1}{6}(\ln 4 - \ln 5) \\ &= \frac{1}{6}\ln\left(\frac{5}{4}\right).\end{aligned}$$

(d) $\displaystyle\iint_R \frac{y}{(1+xy)^2}\,dA$, where $R = \{(x,y) \mid 0 \le x \le 2,\ 0 \le y \le 1\}$.

$$\begin{aligned}\iint_R \frac{y}{(1+xy)^2}\,dA &= \int_0^{1}\int_0^{2} \frac{y}{(1+xy)^2}\,dx\,dy \\ &= \int_0^{1} \left[-\frac{1}{1+xy}\right]_{x=0}^{x=2}\,dy \\ &= \int_0^{1} \left(1 - \frac{1}{1+2y}\right)\,dy \\ &= \left[y - \frac{1}{2}\ln(1+2y)\right]_0^{1} \\ &= 1 - \frac{1}{2}\ln 3.\end{aligned}$$

5. Find the volume of the solid that lies under the hyperbolic paraboloid

$$z = 3y^2 - x^2 + 2$$

and above the rectangle $R = [-1, 1] \times [1, 2]$.

Observe that $z \geq 0$ for all $(x, y) \in R$. Thus, the volume is given by

$$\begin{aligned} V &= \int_{-1}^{1} \int_{1}^{2} (3y^2 - x^2 + 2)\, dy\, dx \\ &= \int_{-1}^{1} \left[y^3 - yx^2 + 2y\right]_1^2 dx \\ &= \int_{-1}^{1} \left[(8 - 2x^2 + 4) - (1 - x^2 + 2)\right] dx \\ &= \int_{-1}^{1} (9 - x^2)\, dx \\ &= \left[9x - \frac{x^3}{3}\right]_{-1}^{1} \\ &= \frac{52}{3}. \end{aligned}$$

6. Find the average value of $f(x, y) = e^{2x-y}$ on the rectangle $R = [0, \ln 3] \times [0, \ln 2]$.

We first evaluate the double integral

$$\begin{aligned} \iint_R e^{2x-y}\, dA &= \int_0^{\ln 3} \int_0^{\ln 2} e^{2x-y}\, dy\, dx \\ &= \int_0^{\ln 3} \left[-e^{2x-y}\right]_{y=0}^{y=\ln 2} dx \\ &= \int_0^{\ln 3} \left(-\frac{1}{2}e^{2x} + e^{2x}\right) dx \\ &= \int_0^{\ln 3} \frac{1}{2}e^{2x}\, dx \\ &= \left[\frac{1}{4}e^{2x}\right]_0^{\ln 3} \\ &= \frac{1}{4}(9 - 1) \\ &= 2. \end{aligned}$$

The area of R is $(\ln 3 - 0)(\ln 2 - 0) = (\ln 3)(\ln 2)$. Thus the average value of f over R is

$$f_{\text{avg}} = \frac{1}{(\ln 3)(\ln 2)} \iint_R e^{2x-y}\, dA = \frac{2}{(\ln 3)(\ln 2)}.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Method of Lagrange Multipliers. To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming that such extrema exist, that f and g are differentiable, and that $\nabla g \neq \mathbf{0}$ on the constraint curve), we find all values x, y, λ such that

$$\nabla f(x, y) = \lambda \nabla g(x, y) \qquad \text{and} \qquad g(x, y) = k.$$

Then evaluate f at each corresponding point (x, y). The largest of these values is the maximum value, and the smallest is the minimum value.

Double Integrals over Rectangular Regions.
If f is continuous on the rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy.$$

If $f(x, y) \geq 0$ on R, then $\iint_R f(x, y)\, dA$ represents the volume of the solid that lies above R and below the surface $z = f(x, y)$.

The Average Value of a Function of Two Variables.
The average value of a function f over a region R of area $A(R)$ is

$$f_{\text{avg}} = \frac{1}{A(R)} \iint_R f(x, y)\, dA,$$

provided that $0 < A(R) < \infty$.

Suggested Textbook Problems
Section 14.8: 1-29
Section 15.1: 1-49, 53-56